

# **UK PATENT APPLICATION**

## **SYSTEM AND METHOD FOR AUTONOMOUS CROSS-DOMAIN CAUSAL DISCOVERY AND BUSINESS GENERATION USING ENSEMBLE PREDICTION AND TEST-DRIVEN DEVELOPMENT**

### **ABSTRACT**

A computer-implemented system for autonomous cross-domain causal discovery and business generation comprising: a data ingestion pipeline collecting behavioural and macroeconomic time-series data from heterogeneous sources; a correlation engine computing pairwise associations using multiple correlation coefficients; a bidirectional Granger causality module determining temporal causal direction via frequency-adaptive lag selection; a three-model ensemble prediction engine combining Granger causality, ordinary least squares regression, and autoregressive integrated moving average forecasts with automatic weight calibration from historical accuracy; a cross-layer enforcement architecture constraining predictions from behavioural signal variables to macroeconomic target variables; an exploitation scoring module ranking opportunities using a weighted formula; an autonomous test-driven development pipeline converting statistical signals into deployed web applications; a deployment module deploying generated applications to cloud infrastructure without human intervention; an outcome measurement subsystem validating predictions and recalibrating model weights; and a build-iterate loop diagnosing failures and spawning corrective builds. The system operates as a closed-loop autonomous business pipeline.